SY B.Tech. – ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PRACTICAL ASSIGNMENT TEMPLATE

1. Student Details

Student Name	Pratik Akrur Pandit
PRN	202501110116
Division	B
Batch	B2
Roll No.	202501110116
Date of Submission	16/10/2026

2. Practical Details

Practical No.	5
Practical Title	Function Types, Friend Class
Date Performed	12/09/2026
Faculty Name	Shruti Gadewar

3. Problem Statement

To study and implement different function types and class-related concepts in C++, including inline functions, static member functions, the this pointer, and friend classes. The practical aims to demonstrate how these concepts are used in object-oriented programming to improve code organization, access class members, and perform operations on objects.

4. Objectives

1. To understand classes, objects, and member functions in C++.
2. To implement inline functions, static member functions, and the this pointer.
3. To understand and demonstrate the use of friend classes to access private members of another class.

5. Expected Learning Outcomes

After completing this practical, the student will be able to:

1. Explain the purpose of classes, objects, and different types of member functions in C++.
2. Implement inline functions, static member functions, and the this pointer in a program.
3. Demonstrate how a friend class can access the private and protected members of another class.

6. Theory / Concept

Classes and Objects: A class is a user-defined data type that combines data members and member functions. An object is an instance of a class used to access its members.

Inline Function: An inline function is declared using the inline keyword. It suggests that the compiler may replace a function call with the function's code to reduce function-call overhead.
Static Member Function: A static member function belongs to the class rather than an individual object. It can directly access only static members of that class and can be called using the class name.
This Pointer: The this pointer refers to the current object of a class. It is useful for distinguishing data members from function parameters with the same name.

Friend Class: A friend class is declared using the friend keyword inside another class. It can access the private and protected members of that class, even though those members are not publicly accessible.

These concepts are important in object-oriented programming because they support code organization, controlled access, and efficient implementation of class-based programs.

7. Algorithm / Pseudocode

START

Step 1: Define a class containing private data members and public member functions.

Step 2: Implement an inline function to perform a simple calculation.

Step 3: Declare a static data member and a static member function to maintain and display shared class information.

Step 4: Use the this pointer to assign values to the current object's data members.

Step 5: Declare a friend class and use its member function to access private data of the original class. Display the results of all operations.

STOP

08. Technology / Tools / Libraries Used

Technology / Tool	Purpose
Programming Language	C++
IDE / Platform	Visual Studio Code
Libraries / Packages	iostream

09. Implementation / GitHub Repository

Particular	Details

GitHub Repository URL	https://github.com/panditpratik/OOP-
Repository Name	**Problem-Solving-Using-OOP-in-Cpp**
Branch Used	**main**
Commit ID / Latest Commit	**7629d9bcebc099e457d54dfa32a1de53bd9ccb9e**
Date of Last Commit	**17/10/2026**

10. Test Cases / Input and Output

Test Case	Input	Expected Output	Actual Output	Status
TC-1	a = 5, b = 3	Sum = 8	Sum = 8	Pass
TC-2	Name = Aman, Age = 60	Name: Aman, Age: 60	Name: Aman, Age: 60	Pass
TC-3	Create two Student objects	Total objects: 2	Total objects: 2	Pass
TC - 4	Balance = 12000	Balance: 12000	Balance: 12000	Pass

11. Results / Observations

Mention the important results obtained during execution.

The program was successfully implemented using a class and object in C++. The member functions were used to accept and display student information. The program correctly displayed the result based on the marks entered.

Screenshots / Graphs / Output:

Practical_assignment_5.cpp > main()

```cpp

#include <iostream>
#include <string>
using namespace std;

// Inline function
inline int add(int a, int b) {
    return a + b;
}

// Class demonstrating this pointer and static members
class Student {
    string name;
    int age;
    static int count;

public:
    Student(string name, int age) {
        this->name = name;
        this->age = age;
        count++;
    }

    void display() {
        cout << "Name: " << this->name << endl;
        cout << "Age: " << this->age << endl;
    }

    static void displayCount() {
        cout << "Total objects: " << count << endl;
    }
};

int Student::count = 0;

// Class demonstrating friend class
class Account {
    int balance;

```

Practical_assignment_5.cpp > main()

```cpp
// Class demonstrating friend class
class Account {
    int balance;

public:
    Account(int balance) {
        this->balance = balance;
    }

    friend class AccountViewer;
};

class AccountViewer {
public:
    void showBalance(Account a) {
        cout << "Balance: " << a.balance << endl;
    }
};

int main() {
    cout << "Inline Function" << endl;
    cout << "Sum: " << add(5, 3) << endl;

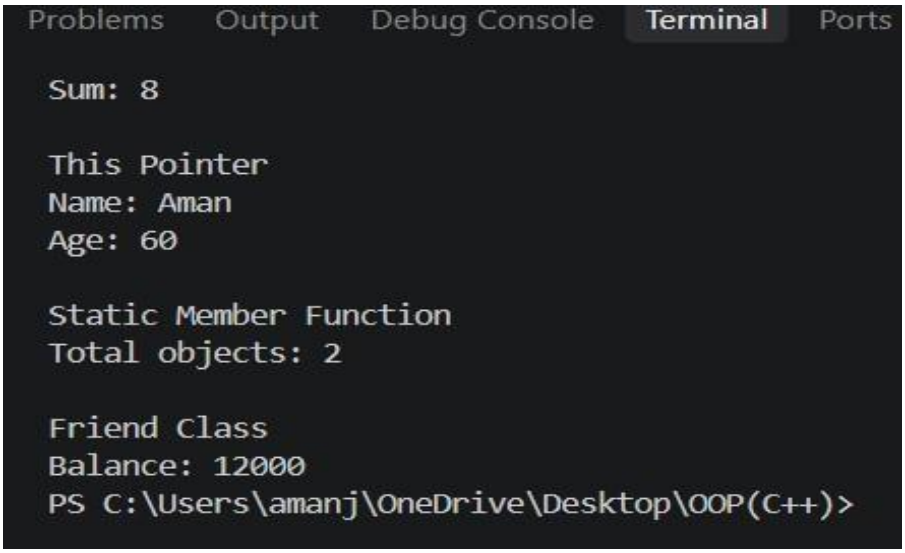
    cout << "\nThis Pointer" << endl;
    Student s1("Aman", 60);
    s1.display();

    cout << "\nStatic Member Function" << endl;
    Student s2("Mayur", 55);
    Student::displayCount();

    cout << "\nFriend Class" << endl;
    Account a(12000);
    AccountViewer viewer;
    viewer.showBalance(a);

    return 0;
}
```

Output :



12. Conclusion

In this practical, different C++ concepts such as inline functions, static member functions, the this pointer, and friend classes were implemented successfully. The program helped in understanding how objects access their members, how static members are shared among objects, and how a friend class can access private data. Thus, the practical improved understanding of object-oriented programming in C++.

13. Assignment Questions

Q1. _____

Answer:

_____

_____

_____

Q2. _____

Answer:

_____

_____

_____

14. Student Declaration

I hereby declare that the above practical work has been completed by me and that the implementation, analysis, and answers are my own work.

Student Name	PRN	Signature

Pratik Akrur Pandit	202501110116	